Lec-48
04/09/26 7:09 PM

# Bit - Masking
0/1 → variable → mask

~(1) = 0
~(0) = 1

&& → logical and
& → Bitwise and
| → " or
~ → " not
^ → " XOR
<< → Left shift
>> → Right shift

a	b	a&b	a|b	a^b
1	1	1	1	0
1	0	0	1	1
0	1	0	1	1
0	0	0	0	0

XOR
0^a = a
a^a = 0
5^5 = 0
101
101
000

101
000
101

a=5
a^b^c^a = c^a^a = c

a^b^c^c
a=5
b=4
c=2
101
011
110
010
101

(2, 3, 1, 3, 2) → ①

Left Shift (<<)

5<<2 → $5 \times 2^2 = 20$
5<<4 → $5 \times 2^4 = 80$
5<<6 → 5×2^6
5<<1 → 5×2^1

a<<b
$= a \times 2^b$

~~0~~0000 101
0000 1010

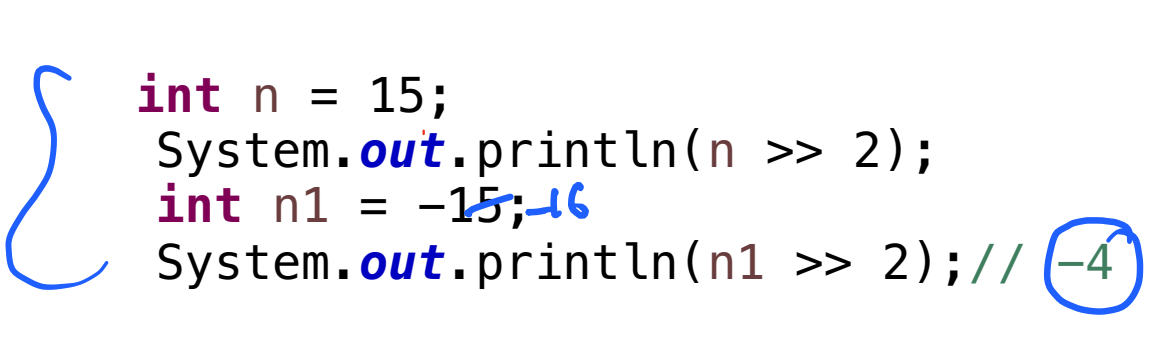
~~0~~000 0101
0000 1010

N = 10
0 000 1010
0 000 0101 → 5

a>>b = $a/2^b$
10>>1 = $10/2^1 = 5$

$-15/2 = -7$
-15>>1 =

-16>>2 = -4
-15>>2 = -4



```
int n = 15;
 System.out.println(n >> 2);
 int n1 = -15;-16
 System.out.println(n1 >> 2);// -4
```

0000 1111
1111 0000
1
①1111 0001

①1 11000
0000 0111
1111
0001000 -8

0 1 1 0 1 0 1 0
000000 1 ① → 0 ✓
8
(n&1) == 0

even
2^{31} ... 2^4 2^3 2^2 2^1 2^0
2^0 → odd
0 → even
1 → odd

2 | 39
2 | 19 1
2 | 9 1
2 | 4 1
2 | 2 0
2 | 1 0
  0 1

100111

if((n%2)==0){
}

18/9 9×2 = 18
(n/2)×2 == n → even
(7/2)×2
3×2 = 6 → odd

64
16

0101010100
43210
4 Rem

1<<4
10
100
1000
10000

(Code) 0101 0100 → mask
0001 0000
000 10000
mask 1<<pos

n = n>>4
(n&1) == 1 → set
→ unset

③ → set only time
0101 0100 → 3

2 | 84 0
2 | 42 0
2 | 21 1
2 | 10 0
2 | 5 1
2 | 2 0
2 | 1 1
  0

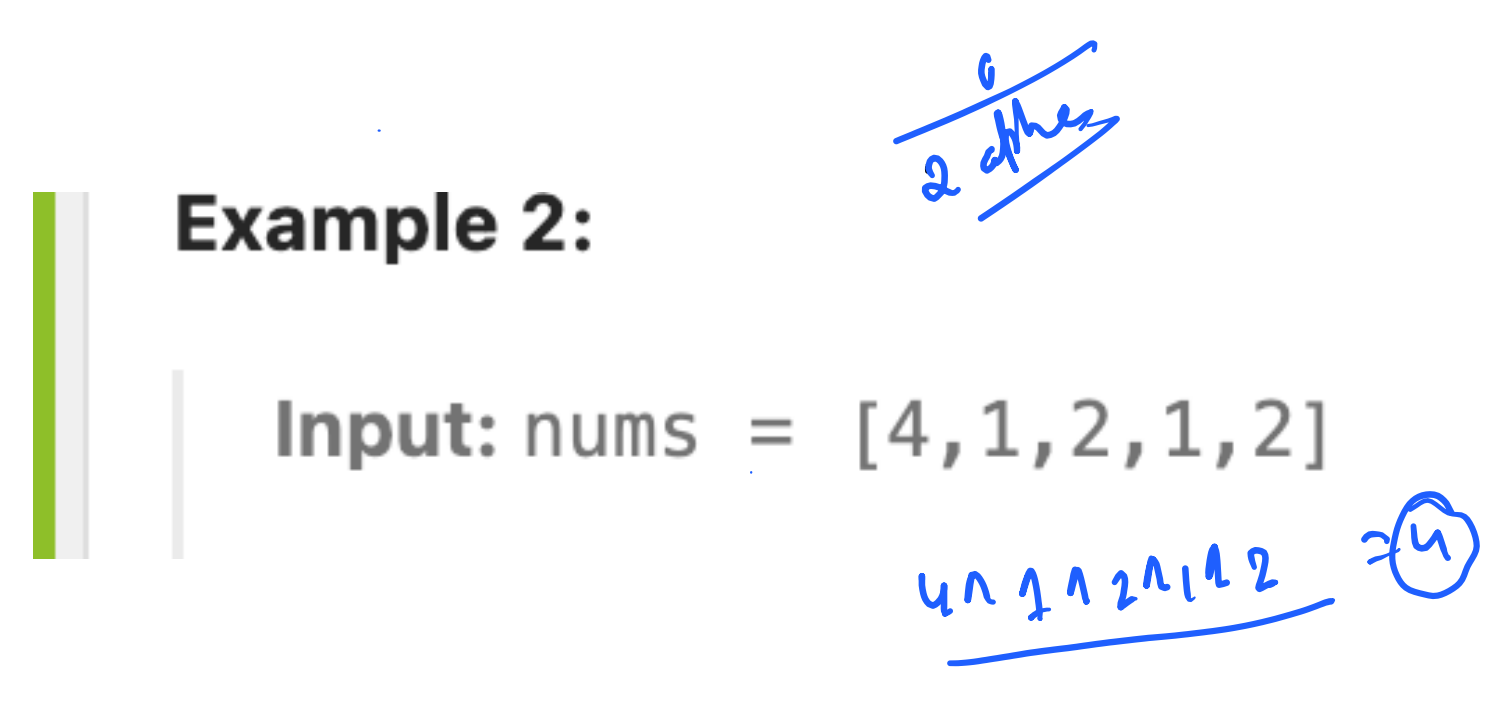
log₂(n)
if((n&1)==1){
c++
}
n = n>>1

while(n>0){
if(n%2==1){
c++
}
n = n/2
3

N → 84 → &
N-1 → 83
0101 0100
0101 0011
0101 0000
0000 1111
0100 0000
& 0011 1111
0 000 0000

while(n>0){
n = (n&(n-1))
c++
3

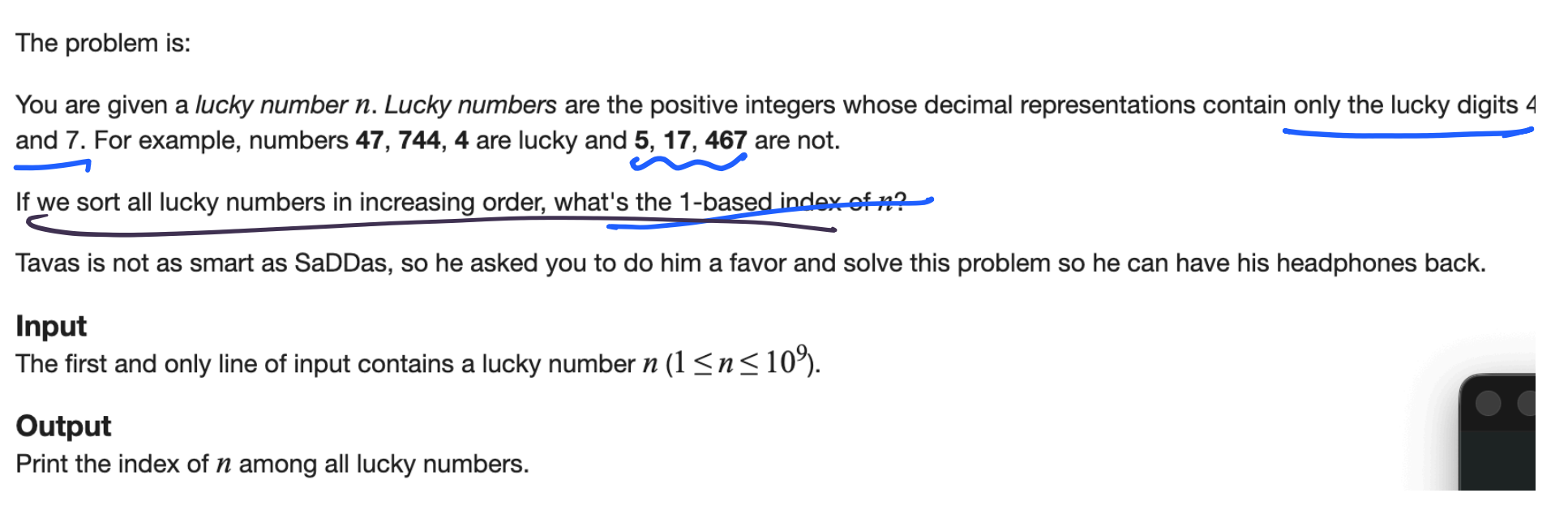
0
2 others



Example 2:

Input: nums = [4,1,2,1,2]

4^1^1^2^1^1^2 = ④



The problem is:

You are given a *lucky number n*. *Lucky numbers* are the positive integers whose decimal representations contain only the lucky digits 4 and 7. For example, numbers **47**, **744**, **4** are lucky and **5**, **17**, **467** are not.

If we sort all lucky numbers in increasing order, what's the 1-based ~~index of n~~?

Tavas is not as smart as SaDDas, so he asked you to do him a favor and solve this problem so he can have his headphones back.

Input
The first and only line of input contains a lucky number n $(1 \le n \le 10^9)$.

Output
Print the index of n among all lucky numbers.

4747474

⑧
4 7 44 47 74 77 444 447 474 477 744 747 - -
1 2 3 4 5 6 7 8 9 10 11 1

4 7 44 47 74 77 444 447 474 477 - - -

2
1 2 3 4 5 6 < 474774
6 digit
2 2

n digit
(n-1) = x
< n digit = y

x+y+1

$x = 2^6 - 2$
$= 62$

$2^1\ 2^2\ 2^3\ 2^4 \ldots \ldots 2^{n-1} = x$

[1<<n] = 1×2^N

$\frac{2[2^{N-1}-1]}{2-1} = 2 \times 2^{N-1} - 2$
$= (2^N - 2) = x$

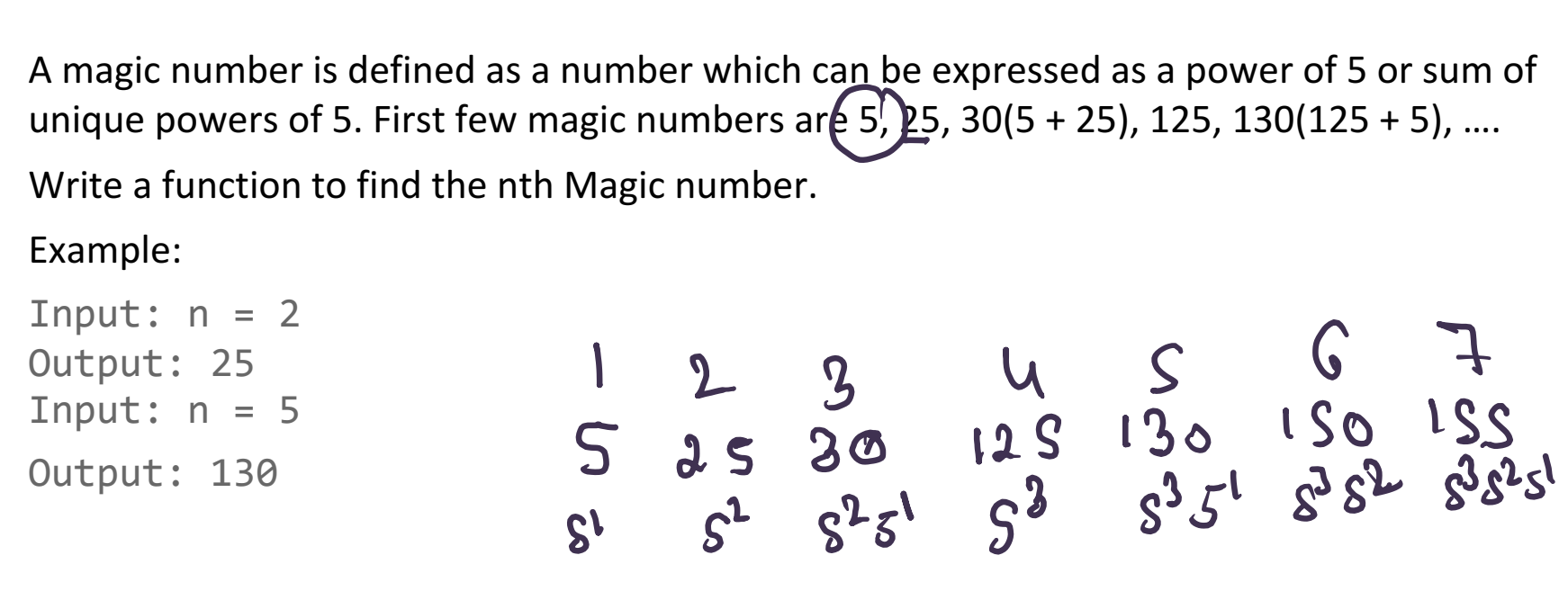
474774
4 - - - -
2^4

$2 + 2^2 + 2^4$

Pos = ~~0~~ ~~1~~ 2 3 4 5 6
$2^1 + 2^2 + 2^4$

474774
4

$2^1\ 2^6\ 2^5\ 2^4\ 2^3\ 2^2\ 2^1\ 2^0$



A magic number is defined as a number which can be expressed as a power of 5 or sum of unique powers of 5. First few magic numbers are 5, 25, 30(5 + 25), 125, 130(125 + 5), ….

Write a function to find the nth Magic number.

Example:

```
Input: n = 2
Output: 25
Input: n = 5
Output: 130
```

1	2	3	4	5	6	7	8	9	10	11
5	25	30	125	130	150	155	625	630	650	655
5^1	5^2	$5^2 5^1$	5^3	$5^3 5^1$	$5^3 5^2$	$5^3 5^2 5^1$	5^4	$5^4 5^1$	$5^4 5^2$	$5^4 5^2 5^1$

5^3

1001
100
10
1 → 0

1001
$5^4 5^3 5^2 5^1$
$5 + 5^4$
630

mul = 5
while(n>0){
if((n&1)==1){
sum = sum+mul
}
mul = mul×5
i >>= 1

3

abc → Recursion
a
ab
b
c
bc
abc
ac

Power Set
pos = 0
abc = 2^3 → 8

	abc
000	-
001	a
010	b
011	ab
100	c
101	ac
110	bc
111	abc

ac

i = 5
101
while(i>0){
if((i&1)!=0){
ans += s.charAt(i)
}
i = i >> 1
pos++